

Assalamu Alaikum. Good morning respected teachers and everyone. I am Dipu Banik. Today I will present our thesis titled, "Analyzing Hypothesis-Only Bias in Bangla and Banglish Natural Language Inference Using Explainable AI."

Introduction

Natural Language Inference, or **NLI**, is an important task in Natural Language Processing.

The goal of NLI is to find the relationship between two sentences: a **Premise** and a **Hypothesis**.

The relationship can be of **three types**:

- Entailment,
- Contradiction,
- and Neutral.

In our research, we study transformer-based models for **Bangla and Banglish NLI** and check whether they really understand the meaning of both sentences.

Motivation and Problem Statement

Bangla and Banglish are widely used in daily life, so it is important to build reliable NLI models for these languages.

A good NLI model should use **both the premise and the hypothesis** to make a prediction.

But sometimes, transformer models depend only on the **hypothesis** or simple shortcut patterns instead of understanding both sentences.

Because of this, a model may get **high accuracy**, but it does not always mean it truly understands the text.

Proposed Solution

To solve this problem, we first prepared and cleaned a **parallel Bangla–Banglish NLI dataset**.

Then, we trained and compared different transformer-based models.

After that, we used **SHAP**, an Explainable AI technique, to find which words in the hypothesis had the biggest effect on the model's prediction.

Finally, we compared the **full-input** and **hypothesis-only** results to check whether the models use real semantic understanding or only shortcut patterns.

Literature Review

We studied several recent research papers on Natural Language Inference.

Most of the previous studies focused on **English datasets**.

Only a few studies worked on **Bangla**, and **no study explored Banglish NLI**.

Also, very few studies used Explainable AI to analyze hypothesis-only bias.

These limitations inspired our research.

Research Gap

From our literature review, we found four main research gaps.

- Most NLI research focuses on **English**, with limited work on **Bangla**.
 - **Banglish NLI has not been explored.**
 - **Hypothesis-only bias has not been studied in Bangla and Banglish NLI.**
 - Existing studies do not show **which hypothesis words** affect hypothesis-only predictions.
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Key Contribution

To fill these gaps, our research makes five main contributions.

- We developed a **parallel Bangla–Banglish NLI dataset**.
- We compared models using **full-input** and **hypothesis-only** settings.
- We analyzed **hypothesis-only bias** in Bangla and Banglish.
- We used **SHAP** to identify the important hypothesis words.
- Finally, we checked whether the models make predictions based on **real understanding** or **hypothesis-only bias**.

Thank you. Now I would like to invite my teammate to continue with the methodology section.

Introduction

Assalamu Alaikum. Good morning respected teachers and everyone. I am Dipu Banik. Today I am going to present our thesis titled, "Analyzing Hypothesis-Only Bias in Bangla and Banglish Natural Language Inference Using Explainable AI."

Natural Language Inference, or **NLI**, is a fundamental task in Natural Language Processing. Its goal is to determine the semantic relationship between two sentences: a **Premise** and a **Hypothesis**.

The relationship is classified into **three categories**:

- Entailment,
- Contradiction,
- and Neutral.

In our research, we evaluate transformer-based models for **Bangla and Banglish NLI** and investigate whether these models truly understand the meaning of both sentences.

Motivation and Problem Statement

Bangla and Banglish are widely used in daily communication, so building reliable NLI systems for these languages is important.

Ideally, an NLI model should consider **both the premise and the hypothesis** before making a prediction.

However, transformer models sometimes rely on **shortcut patterns** or only the **hypothesis**, instead of understanding the relationship between both sentences.

As a result, a model may achieve **high accuracy**, but that does not necessarily mean it has learned true semantic understanding.

Proposed Solution

To address this problem, we first prepared and preprocessed a **parallel Bangla–Banglish NLI dataset**.

Then, we trained and compared multiple transformer-based NLI models.

Next, we applied **SHAP**, an Explainable AI technique, to identify which hypothesis words most influenced the model's predictions in the **hypothesis-only setting**.

Finally, we compared the results of the **full-input** and **hypothesis-only** settings to analyze the presence of hypothesis-only bias and determine whether the models rely on semantic understanding or shortcut patterns.

Literature Review

We reviewed several recent studies on Natural Language Inference.

Most previous works focused on **English datasets**, while only a few considered **Bangla**.

Although some studies explored explainability or hypothesis-only bias, they mainly used English datasets such as **SNLI** and **MNLI**.

Very few studies investigated Bangla, and **none explored Banglish NLI** or analyzed hypothesis-only bias using Explainable AI.

These limitations motivated our research.

Research Gap

From our literature review, we identified four major research gaps.

- First, most NLI research focuses on **English**, with limited work on **Bangla**.
 - Second, **Banglish NLI has not been explored**.
 - Third, **hypothesis-only bias has not been analyzed for Bangla and Banglish NLI**.
 - Finally, existing studies do not identify **which hypothesis words** influence hypothesis-only predictions.
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Key Contribution

To address these gaps, our research makes five main contributions.

- We developed a **parallel Bangla–Banglish NLI dataset**.
- We compared transformer models under **full-input** and **hypothesis-only** settings.
- We analyzed **hypothesis-only bias** in Bangla and Banglish.
- We applied **SHAP** to identify the hypothesis words influencing predictions.
- Finally, we examined whether the models make predictions based on **true semantic understanding** or **hypothesis-only bias**.

Thank you. Now I would like to invite my teammate to continue with the methodology section.